

The accessibility engineer that fixes your code. And proves it.

An agentic workflow that repairs WCAG violations in real website source - then verifies every fix with deterministic re-scans, content-integrity checks and localized visual diffs, rolls back anything destructive, and routes judgment calls to a human with drafted fixes. The output: a fixed site and evidence a business owner can hand to their lawyer.

134 / 134

seeded WCAG violations
remediated & independently
verified

0

hallucinated image descriptions
(the baseline shipped 8 across
two runs)

32

scanner-invisible issues found &
drafted for human review

~\$0.50

measured compute per site - vs
\$350-550/page manual
remediation

A lawsuit, not a lint warning

The typical defendant in a web-accessibility lawsuit is not a tech company. It's a bakery, a dental clinic, a small law firm - a site built by a freelancer in 2019 that nobody has touched since. The owner discovers the problem when the demand letter arrives.

5,114

US web accessibility lawsuits in 2025 - a record (UsableNet year-end report)

64%

of 2025 defendants had revenue under \$25M - small businesses

95.9%

of the top 1M homepages fail WCAG; 56 errors per page on average (WebAIM 2026)

Jun 2025

European Accessibility Act in force - fines to €100k (DE) or 5% of turnover (IT)

Behind the legal risk sits the actual point: ~87 million people in the EU alone live with a disability. For them a missing form label isn't a compliance item - it's a door that doesn't open.

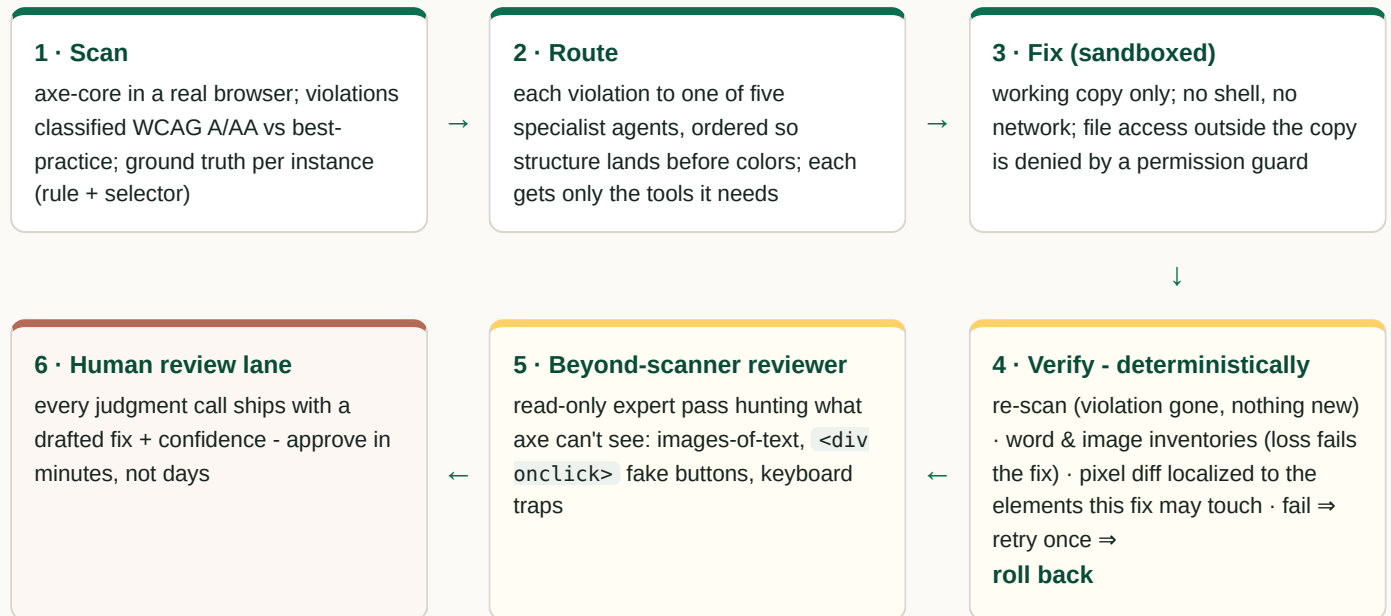
Why today's options fail this user

	WHAT IT DOES	WHY IT FAILS
Overlay widgets	masks issues in a JS layer - code untouched	FTC fined accessiBe \$1M (2025) over compliance claims; ~25% of sued companies had a widget installed when sued; 1,031 practitioners signed the Overlay Fact Sheet against them
Scanners	finds issues, fixes nothing	the report is homework; automation detects ~57% of issues by volume (Deque) and none of the fixes
Consultants	fixes properly	\$1,250-2,750 audit + \$350-550 per page remediation, weeks of lead time - priced for enterprises
Paste into a chatbot	sometimes brilliant, sometimes destructive	we measured it: same clinic page, two runs - one hid the clinic's booking phone number from blind users, one didn't. Nothing tells you which run you got.

The bottleneck in 2026 is not finding problems - it's not even fixing most of them. It's knowing, with evidence, that they were fixed and nothing was silently broken - at a price a bakery can pay.

Agents with a deterministic spine

Language models supply judgment. Everything checkable is checked by machinery the model cannot argue with - and every design choice answers a *documented* failure mode of LLM accessibility repair (research mapping in `docs/research/prior-art.md`).



The three principles

- **Models judge; tools know.** Contrast fixes come from WCAG luminance math that finds the closest compliant color *preserving the brand hue*. Alt text may only be written after the agent has rendered and **looked at** the image. Perception precedes assertion - so the hallucination class is eliminated, not just caught.
- **The verifier is not the fixer.** Acceptance is decided by axe re-scans and content inventories; failed fixes roll back automatically and join the human queue.
- **Honesty is the product.** Automation covers ~57% of issues by volume. StepFree claims verified fixes for the detectable layer and routes the rest to humans - the claim discipline the FTC fined the market leader for lacking.

Memory, skills, orchestration - used purposefully

- **Memory:** a conventions ledger - the first agent to resolve a color pair or set an alt style records it; every later fixer on every page reuses it. One accessible palette per site, not one per page.
- **Skills:** five per-domain knowledge files encode WCAG craft (e.g. "no ARIA beats bad ARIA - prefer removing it"; "an image inside a link describes the destination").
- **Orchestration:** a deterministic pipeline; every changelog stage is a replayable flag (`--stage v1...final`), and every agent run is captured as a full trajectory.

The finding we didn't expect - reported plainly

Corpus: 12 synthetic small-business sites (13 pages), **134 scanner-verified WCAG A/AA violations** mirroring the WebAIM distribution, ground truth locked before any run. Baseline = the same model (cLaude-sonnet-5), one direct prompt. The evaluator is independent of the system under test: it re-scans and re-screenshots every page itself.

On small static pages, the one-shot baseline remediated **134/134** scanner-detectable instances with zero regressions - frontier-model capability on the *scanner-visible* layer is saturated, and we say so. The layers that decide whether a real person can use the site are where the systems separate:

	BASELINE - ONE DIRECT PROMPT	STEPFREE - FINAL
WCAG A/AA instances remediated & verified	134/134	134/134 · 13/13 pages shippable
Alt-text truthfulness (0-2, vision-judged by Claude Opus)	1.53 / 1.37 across two runs - 4 hallucinated or destructive alts each run	1.80 (v3: 1.83) - no fabricated content (all four deductions enumerated & defended in EVALUATION.md)
Informative content silenced with alt=""	1 of 2 runs - including the clinic's booking phone number	0 - the image agent must look before writing
Scanner-invisible issues surfaced for humans	0 (by definition)	32, each with a drafted fix
Evidence produced	none - ship on faith	audit report + per-fix verification + full trajectories
Measured compute, whole corpus	\$2.03	\$5.78 (~\$0.48/site)

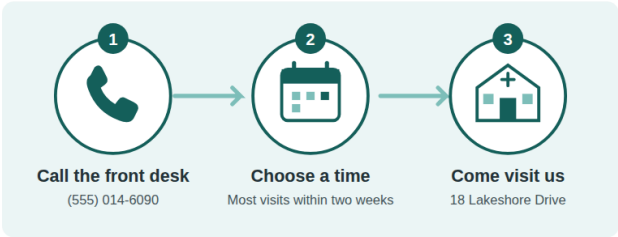


Exhibit A. The baseline resolved this image's missing-alt violation by marking it decorative (alt="") - hiding the phone number and address from screen-reader users. The scanner scored the page as *improved* (11 → 0). StepFree's vision-grounded agent read the image and wrote alt text carrying the steps; run two of the baseline described it but dropped the phone number. Same input, coin-flip outcomes - unless something checks.

Other baseline exhibits (judge's reasoning on file)

- "Two wine glasses beside an uncorked bottle" - image shows **one** glass and a **sealed** bottle.
- "Officials break ground with ceremonial shovels..." - an entirely invented ceremony; the image is a building.
- An image-of-text hero given alt that doesn't match the words in the image (WCAG 1.4.5).

Economics per 13-page site

	HUMAN TIME	COST
Manual remediation	~28 h	\$4,550-7,150 (at \$350-550/page)
Baseline + due-diligence QA	~4.3 h	\$2 + QA labor
StepFree	~1 h (review queue + report)	\$5.78 (~\$0.48/site) compute

Assumptions stated in docs/research/cost-model.md; primary metric unaffected by them.

Widiflour Bakery

Share this page

Print this page

Sign in

Baked before sunrise, gone by noon.

Historically, powdered flours and butter pastries, made by hand in small batches since 1914.

See what's baking





Osteria del Ponte

Località della Spona, 50100 Sesto San Giovanni - 021 27882

www.osteriadel Ponte.it

Antipasti *Small plates to open the evening, made to sharing*

Burrata di Puglia 14
Cream filled burrata, Marinated cherry tomatoes, basil oil, grilled vegetables

Carpaccio di Manzo 18
Thin raw beef steaks, dressed parmesan, orange, lemon and olive pressed olive oil

Assortito di Formaggi 12
Crisp soft cheese slices with black truffle and a no-bake brioche honey cake





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Introducing TorqueWORKS, a nationwide independent garage.

From check engine lights to full brake jobs, our technicians explain what your car needs, what can wait, and what's a total cost – before we pick up your wrench.



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Some key appointments already available – call before 10 AM.

SERVICES & PRICING





An advertisement for Ember & Oak Studio. At the top, the brand name "Ember & Oak" is in a serif font, followed by "Studio" in a smaller font. To the right, the text "SHOP" and "OUR STORY" are in a sans-serif font. Below this, the phrase "HAND-POURED IN SMALL BATCHES" is centered in a serif font. The main headline "Candles for slow evenings" is in a large, elegant serif font. A paragraph of text follows: "Every Ember & Oak candle is poured by hand in our Hudson Valley workshop from 100% soy wax, cotton wicks, and fragrance oils we blend ourselves. Clean burn, honest ingredients, no shortcuts." Below this text is a dark rectangular button with the text "SHOP FOR COLLECTION" in white, all-caps serif font. At the bottom, the word "Bestsellers" is in a serif font. Below the text are two images of candles. The left image shows a dark, cylindrical candle in a dark holder. The right image shows a tall, light-colored candle in a light-colored holder, with a small heart icon above it. The background of the advertisement is a light, textured color.



Ember & Oak

SHOPS OUR STORY

OUR STORY

Two kitchens, one drafty barn, and a lot of wax

Ember & Oak started in 2021 when Maya Okady melted down the staks of every candle in her apartment because the candles *stunk* every time — and decided the brand she made smelled better than anything she'd ever seen. Her husband Sam took her her pouring to get a bit of a hobby and a job done.

Four years later we are about to launch candles a week in a converted garage in the Flatiron City, all in batches and enough that every single one is packed, stored, and labeled by a person whose name you could learn.

THE WORKSHOP

Where the pouring happens



Second Chance Animal Rescue

Adopt

Get Involved

Contact

Every animal deserves a second chance.

There's a volunteer on rescue in Cedar Falls that has placed more than 1,400 dogs, cats, and small animals into permanent homes since 2011.

Every adoptable animal lives in a happy home – not in a cage – so we can tell you exactly who your bringing home.

Meet the animals.

Waiting for a home

All adoptable animals: dogs, cats, puppies, kittens, and two weeks of follow-up support. Adoption fees: dogs \$175, cats \$95, small animals \$85.



The image shows a web browser window displaying the homepage of 'Wines Lane'. The navigation bar at the top includes 'Services', 'Stylets', 'Visit', and a 'Book now' button with a calendar icon. The main content area features a large heading 'Hair that feels like you, only better.' followed by a paragraph: 'A small studio on Wines Lane where appointments never feel rushed. The coffee is always fresh, and your stylist actually remembers what you talked about last time.' Below the text is a button labeled 'Book an appointment'. To the right of the text is a stylized illustration of a modern interior space with a round mirror, a potted plant, and a pink chair.

[illegible]

A screenshot of the 'HOLD UP' event page on The Foundry website. The page has a dark background with a light blue and white geometric pattern. At the top, the 'THE FOUNDRY' logo is on the left, and navigation links 'Speakers', 'What's included', and 'Book' are on the right. The main heading 'HOLD UP' is in large, bold, white capital letters. Below it, the subtitle 'an *the Foundry* event' is in a smaller, italicized font. A paragraph of text describes the event: 'A 1912 men's frothy on the coast, stripped to its bones and rebuilt for celebrations, Expanded links, forty-foot trusses, and staff who have seen every kind of party go right.' At the bottom, there are two orange buttons: 'CHECK AVAILABILITY' and 'TAKE THE VIRTUAL TOUR'. A small orange triangle icon is centered below the buttons. The bottom of the image shows a portion of a browser window with a blue address bar and a yellow navigation bar.



Lakeside Family Medicine
 Primary Care

[Services](#)
[Providers](#)
[New patients](#)
[Contact](#)
[Patient portal](#)

Care for every age, right by the lake.

From random checkups to Medicare wellness visits, our small team has looked after Fairview families for twenty-five years – often three generations of them in the same afternoon.

[Request an appointment](#)
[\(555\) 616-6090](#)

Same-day sick visits every weekday morning.



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Six measured stages, one removed experiment

STAGE	WHAT CHANGED & WHY	EVIDENCE (SAME EVAL, ALL 134 INSTANCES)	DECISION
baseline	one direct prompt - what a busy developer does today	134/134 on the scanner layer, but alt truthfulness 1.53/1.37 with 4 hallucinations per run; booking infographic silenced in run 1	the problem moved from capability to groundedness & trust
v1	basic agent: axe report + file tools	134/134 · \$1.48 - cheaper than baseline: targeted edits beat full-page rewrites	kept as chassis
v2	deterministic verify loop (re-scan diff, bounded retry)	134/134; retries fired 0 times on-corpus	kept - converts "probably fine" into "proven"
v3	5 specialists + knowledge files + contrast math + mandatory image viewing	alt truthfulness 1.83, zero hallucinations ; silenced infographic 0 → 2	kept - the load-bearing change
v4	integrity guardrails: word/image inventories, localized pixel diff, auto-rollback	134/134, 0 rollbacks needed - the reason "0 pages damaged" is a fact, not an assumption	kept - quiet gates still testify
final	conventions memory + beyond-scanner review lane + evidence report	134/134 · 32 review items incl. all 5 planted scanner-invisible traps (5/5 recall) · \$5.78 corpus	the product

The experiment we removed: a strict visual-diff gate

"Reject any fix changing >0.5% of pixels" sounds safe. A controlled, LLM-free experiment (`stepfree/eval/pixel-vs-harm.mjs`) shows it's broken both ways:

EDIT	VIEWPORT PIXELS CHANGED	WORDS LOST
Contrast recolor (legitimate)	0.27%	0
Entire menu item deleted (destructive)	1.32%	13
Naive heading retag (legitimate)	1.93%	0

Pixel area measures reflow, not harm - any gate loose enough to admit the legitimate retag admits the deletion. Removed as a gate, kept as a localized advisory; the word-inventory check orders harm correctly.

Hot take. We built verification armor expecting to catch a clumsy model - and the model never stumbled where we watched. It failed by *asserting what it had no way to know*: describing images it never saw, silencing an infographic with a phone number in it. In 2026, agentic scaffolding isn't primarily there to make models capable - it's there to make failures visible, bounded, and impossible to ship silently. Even our tooling agreed: shown an unviewable image, our grounded judge said *"I cannot honestly score this"* - the ungrounded baseline invented a croissant. **Groundedness is an architecture property, not a model property. Capability saturated; trust didn't. Sell trust.**

Run it yourself in an afternoon

Clean-environment reproduction

```
git clone github.com/shil720/micro1-hackathon
cd stepfree && npm install && npx playwright
install chromium
export ANTHROPIC_API_KEY=...
node stepfree/eval/selftest.mjs #
deterministic core
node stepfree/eval/run-eval.mjs verify-
fixtures # lock ground truth
node stepfree/src/cli.mjs fix fixtures/08-
shelter --out runs/demo
bash stepfree/eval/run-all-stages.sh 3 # full
evaluation
```

- Node 22 · Playwright 1.62 · axe-core (via @axe-core/playwright 4.13) · Claude Agent SDK
- Full ladder ≈ 2.5-4 h, ≈ \$18 of API compute; single-site demo ≈ 6 min, ≈ \$1
- All data synthetic & in-repo; scanner + integrity checks fully deterministic; the evaluator refuses to run if fixture ground truth drifts

Deliverables in the repository

- **README.md** - problem, user, solution, results
- **CHANGELOG.md** - every iteration with evidence
- **EVALUATION.md** - metrics defined up front, full tables, limitations
- **REPRODUCTION.md** - this page, expanded
- **trajectories/** - every agent type, readable + raw JSONL
- **reports/** - a generated audit report (it scans 0/0 itself)

Why this is a company, not a demo

- **Wedge:** \$149 one-time "fix & prove" per site · \$49/mo monitoring · \$499/mo agency tier (20 sites). Measured COGS ≈ \$0.50/site → >90% gross margin.
- **First buyers:** web agencies - thousands of 2019-era SMB sites, already fielding the demand-letter call.
- **Market:** accessibility software → ~\$1B by 2030; incumbents at \$40-50M ARR prove willingness to pay - for products that don't fix code. FTC action + 1,000+ suits/yr against widget users are the tailwind for "actually fix it".
- **Moat:** anyone can prompt a model to edit HTML. The verification stack, evidence trail, and honesty posture are what a stranger can trust with production - and what survives a legal letter.

Safety & ground rules

- All edits in a sandboxed working copy; deploying is a human act. Agents: no shell, no network, path-guarded file access.
- Human reviewer in the loop for every judgment call, with drafted fixes.
- Synthetic data only; no real business named; every claim traces to a source or a JSON artifact in the repo.
- Built openly with the tools it evaluates: Claude agents did the heavy lifting under Shivam's direction; every number comes from the deterministic harness.

Curb cuts were built for wheelchairs - everyone with a stroller or a suitcase uses them. Fix the code, and everybody walks in. **This is StepFree.**